Limbaje Formale și Automate
Seminarul 11 – MTC

Mașina Turing Compusă (MTC) = o compunere de MT elementare

MT elementare: $R, L, R_x, L_x, R_{!x}, L_{!x}$ – nu modifică banda

$X \leftarrow$ modifică banda, $x \in \Sigma$

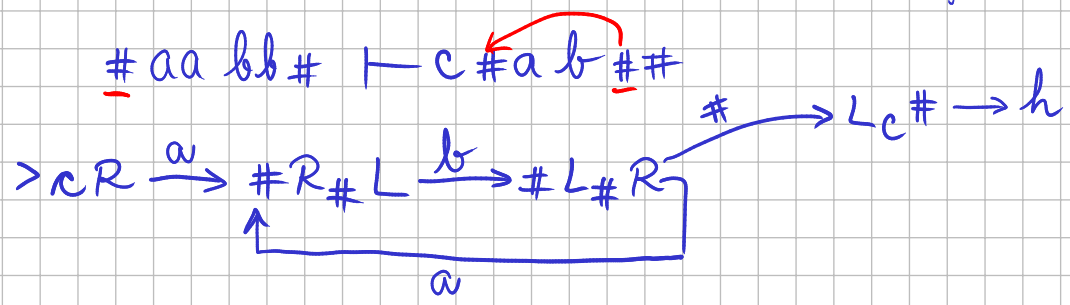
$M_0 \xrightarrow{x} M_i M_j$ (mă aștept să găsesc x pe bandă)

$M_0 \longrightarrow M_i M_j \equiv M_0 M_i M_j$

$M_0 \overset{X}{\uparrow} M_i M_j$
↳ scrie x pe bandă

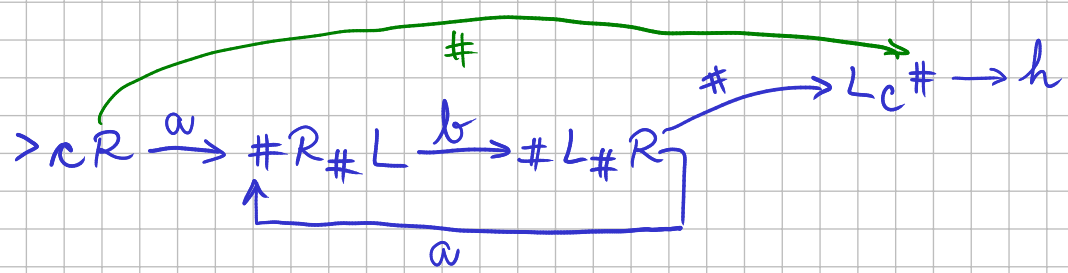
1. MTC care acceptă $L = \{a^n b^n \mid n > 0\}$

$\#aabb\# \vdash c\#ab\#\#$

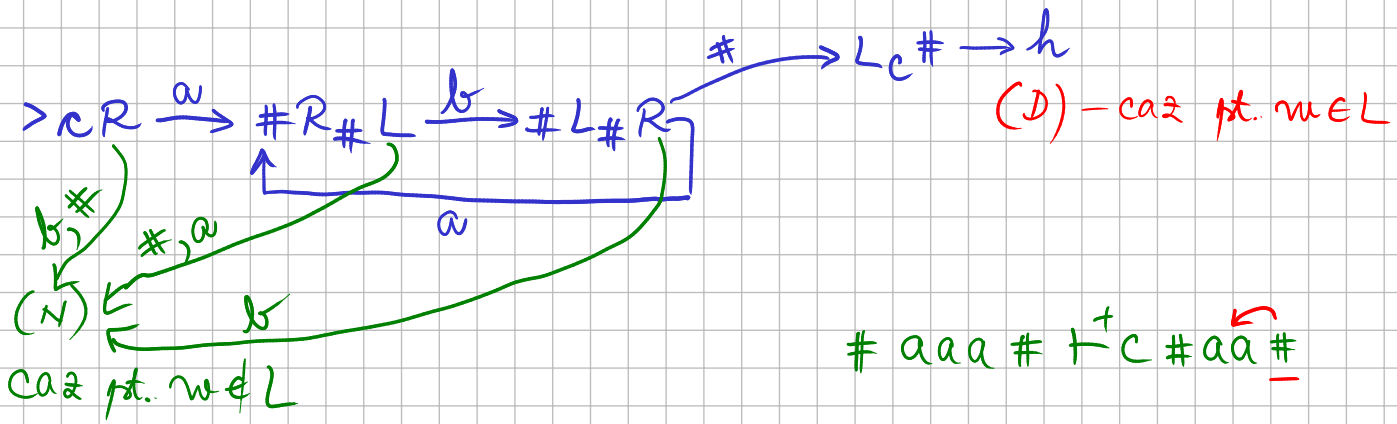


Pt. $L_1 = \{a^n b^n \mid n \geq 0\}$

$\#\# \ldots \vdash c\# \vdash c\#$



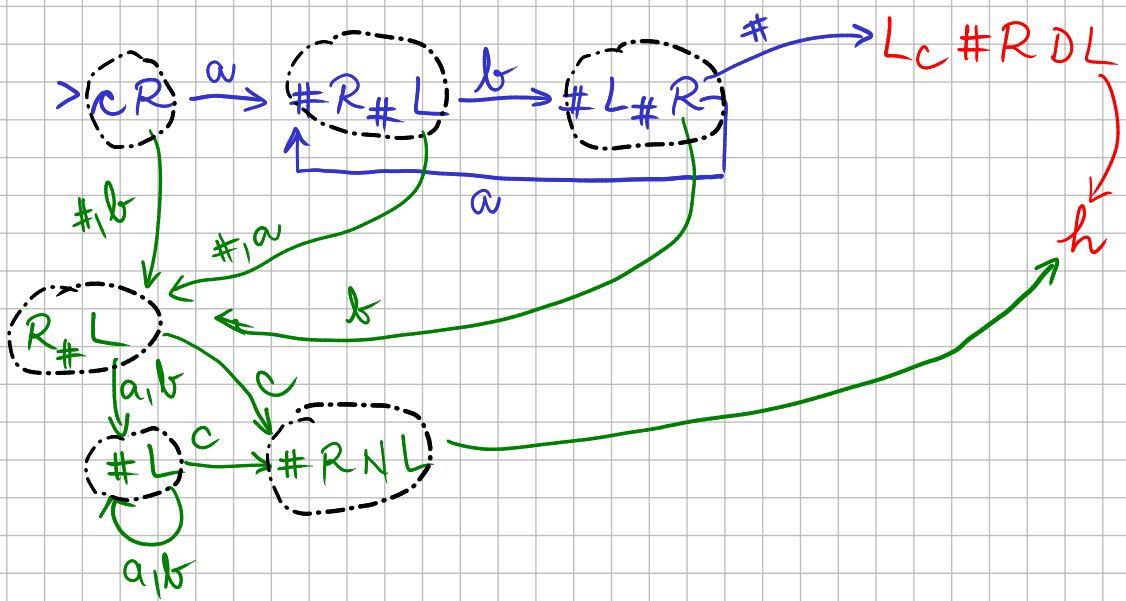
2. MTC care decide $L = \{a^n b^n \mid n > 0\}$



(D) – caz pt. $w \in L$

a a a # ⊢⁺ c # a a

a b b # ⊢* c # b #



3. MTC care acceptă $L = \{a^i b^j c^k \mid i = j \text{ sau } j = k\}$

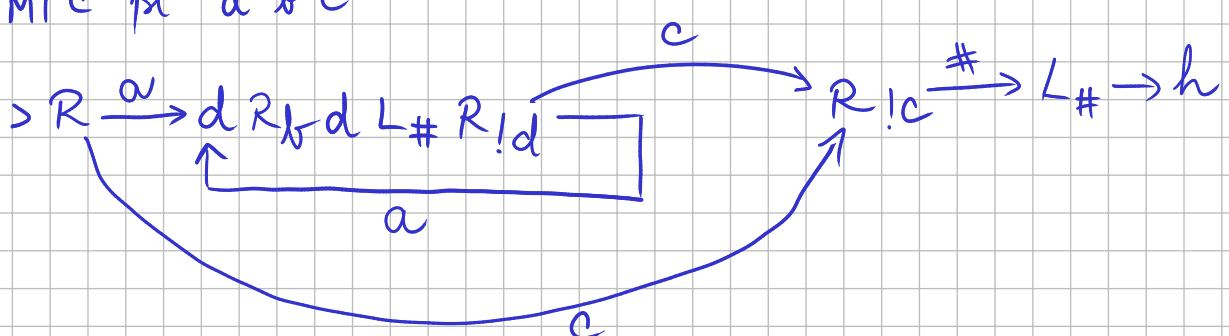
$a^i b^j c^k \rightarrow a^i b^i c^k$

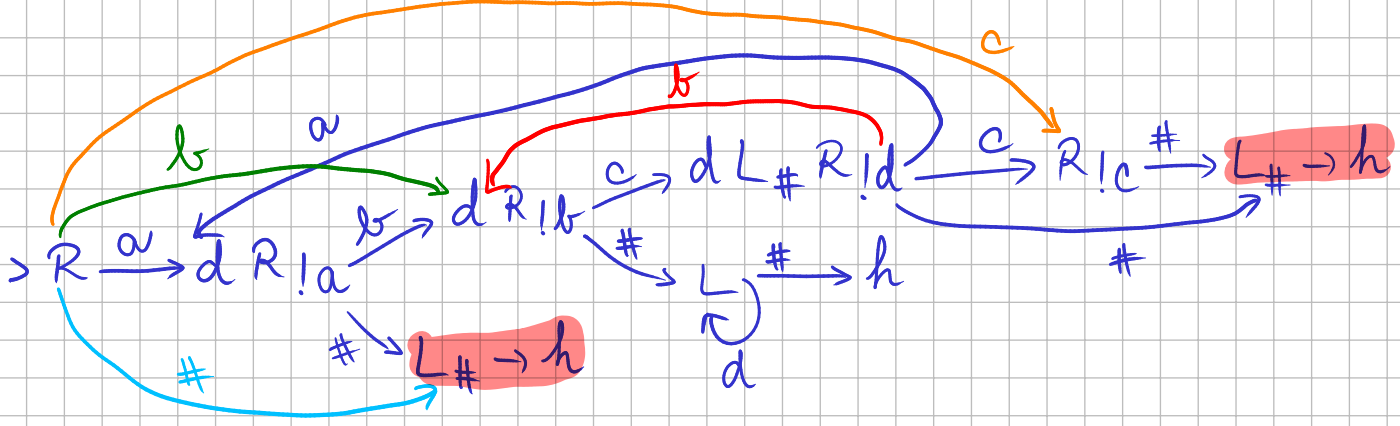
$a^i b^j c^k \rightarrow a^i b^j c^j$

a a b b c c c

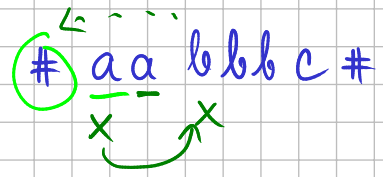
d a d b c c c

MTC pt $a^i b^i c^k$





MTC care acceptă $L = \{w \in \{a,b,c\}^* \mid \#_a(w) + \#_c(w) = \#_b(w)\}$



#bbb#

#abb#

#xxb#

